

A Distribution of Possibilities

Project: Verified Study Buddy

1. Your mission

Your learning target

I can check whether a next-token table is a valid probability distribution.

I can calculate a two-step sequence probability using conditional probabilities.

I can explain why token probability does not measure factual truth.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 2: A Distribution of Possibilities

A generator assigns different likelihoods to possible continuations.



CREATOR MISSION

Run a weighted-word simulation and compare repeated outputs.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why must token counts name the vocabulary and special-token convention?

Readiness for Embeddings and position

Do adjacent vocabulary IDs prove that their tokens have similar meanings?

My readiness explanation

This week's end goal

By the end, I can multiply conditional probabilities to compare paths instead of only first tokens.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use a distribution of possibilities to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A constructed model uses prefix “Status:” and vocabulary alternatives ready, busy and unknown. Its next-token table is ready 0.5, busy 0.3, unknown 0.2. All entries are nonnegative and sum to 1. A probability belongs to a token under this prefix and model, not to that token in every situation. An omitted vocabulary alternative would make the interpretation incomplete unless its probability were explicitly zero.

A table containing 0.6,0.3,0.3 sums to 1.2 and is not already a probability distribution. If the values are nonnegative weights and the rule says normalise, divide each by 1.2. Do not silently treat weights as probabilities. Negative weights cannot be repaired by this simple normalisation rule.

Apply the connected idea: Embeddings and position

Use a two-token vocabulary: ID 1 means red with embedding [1,0]; ID 2 means blue with embedding [0,1]. This lookup table is constructed, not trained. The input red blue has embedding rows [1,0] and [0,1], so its shape is two sequence positions by two components. Reversing the text reverses the rows. ID 2 is not “twice as blue” as ID 1 is red.

A model needs a way to use order. A collection of vectors without any order-sensitive procedure may lose that distinction. We introduce one explicit toy scheme: add position vector [0,0] at position 0 and [2,0] at position 1. These invented values teach arithmetic; they are not a recommended production encoding.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Run a weighted-word simulation and compare repeated outputs.



SYSTEM

A generator assigns different likelihoods to possible continuations.



OUT

EVIDENCE

Run a weighted-word simulation and compare repeated outputs.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Expanded prefix · End probability · Today probability

Expanded prefix: Status: ready

End probability: 0.8

Today probability: 0.2

Expanded prefix: Status: busy

End probability: 0.5

Today probability: 0.5

Expanded prefix: Status: unknown

End probability: 1

Today probability: 0

The probability of ready followed by end is $0.5 \cdot 0.8 = 0.4$. Ready followed by today is $0.5 \cdot 0.2 = 0.1$. Busy followed by end is $0.3 \cdot 0.5 = 0.15$, and busy followed by today is 0.15. Unknown followed by end is 0.2 and unknown followed by today is 0. Together these six two-step paths sum to 1. These products are chain-rule calculations; they do not assume the second token is independent of the first.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Ready probability 0.5

Procedure: End given ready 0.8

Permitted conclusion: Joint path probability 0.4

Plan or input: Busy probability 0.3

Procedure: End given busy 0.5

Permitted conclusion: Joint path probability 0.15

Plan or input: Unknown probability 0.2

Procedure: End given unknown 1

Permitted conclusion: Joint path probability 0.2

The second probability uses the expanded prefix on that branch.

Apply the connected idea: Embeddings and position

Sequence · Position 0 representation · Position 1 representation

Sequence: red blue

Position 0 representation: $[1,0] + [0,0] = [1,0]$

Position 1 representation: $[0,1] + [2,0] = [2,1]$

Sequence: blue red

Position 0 representation: $[0,1] + [0,0] = [0,1]$

Position 1 representation: $[1,0] + [2,0] = [3,0]$

The same red token now has $[1,0]$ at position 0 and $[3,0]$ at position 1. Both token identity and position affect the represented row. Addition requires equal widths; adding a three-component position vector to a two-component embedding violates the input contract. Record the position numbering and embedding-table version.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: ID 1 at position 0

Procedure: $[1,0] + [0,0]$

Permitted conclusion: Representation $[1,0]$

Plan or input: ID 1 at position 1

Procedure: $[1,0] + [2,0]$

Permitted conclusion: Representation $[3,0]$

Plan or input: Two different sequences

Procedure: Inspect ordered rows

Permitted conclusion: Do not infer meaning from ID gaps

The position contribution changes a token representation according to its location.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Prediction: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Reason: Changing one conditional distribution changes its path products while the first-token probabilities remain fixed.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

LIKELIHOOD

relative weight assigned to an outcome

3

SAMPLE

choose one result using the weights

4

CONTINUATION

next token or piece of output

Word · My own example · Not this / boundary

Word: distribution

My own example: A description of how values or probabilities are spread across possible outcomes.

Not this / boundary: _____

Word: likelihood

My own example: In this introductory context, how plausible an outcome is under a model; in statistics, likelihood has a more specific parameter-based meaning.

Not this / boundary: _____

Word: sample

My own example: A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.

Not this / boundary: _____

Word: continuation

My own example: Text that follows an existing sequence.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Probability distribution

Meaning: Nonnegative probabilities over stated alternatives that sum to one

Word: Conditional probability

Meaning: A probability evaluated given specified information

Word: Sequence probability

Meaning: The product of successive conditional probabilities under an autoregressive model

Word: Probability mass

Meaning: The probability assigned to an alternative or a set of alternatives

Terms for Embeddings and position

Term · Meaning

Term: Embedding

Meaning: A vector associated with a token under a specified model

Term: Position vector

Meaning: A vector carrying information about a sequence position

Term: Representation width

Meaning: The number of components in each vector

Term: Elementwise addition

Meaning: Adding corresponding vector components

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 2 • WORKED EVIDENCE

- 1 Possible next words have weights: clear 5, useful 3, magical 1.
- 2 Nine tokens in the spinner represent those relative weights.
- 3 Repeated sampling can produce different sequences.
- 4 One sample does not reveal the whole distribution or guarantee truth.

1. Possible next words have weights: clear 5, useful 3, magical 1.
2. Nine tokens in the spinner represent those relative weights.
3. Repeated sampling can produce different sequences.
4. One sample does not reveal the whole distribution or guarantee truth.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Greedy decoding chooses the largest next-token probability at each step, with an explicit tie rule if needed. It does not always maximise the probability of a complete sequence. Counterexample: first-step $A=0.6$ and $B=0.4$. Given A , $\text{end}=0.51$ and $x=0.49$; given B , $\text{end}=1$. Greedy produces A, end with probability 0.306 . The path B, end has probability 0.4 , larger than 0.306 . This comparison concerns exactly two-step paths; longer completion probabilities require the later conditionals too.

A high path probability measures the model's preference over these specified paths. It is not evidence that a status claim matches a live observation. Format checks and source checks remain separate.

Apply the connected idea: Embeddings and position

The elementwise sum of the represented rows is $[3, 1]$ for both red blue and blue red. This aggregate loses the row-order distinction despite the position additions. A later computation must actually use the represented rows appropriately; merely adding position vectors does not guarantee that every summary preserves order.

A position scheme may use learned values or a deterministic construction. The original Transformer added positional information to embeddings. Modern architectures can use other order-sensitive schemes. Our two-vector example demonstrates one possible mechanism and one information-loss failure, not all architecture choices.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: A Distribution of Possibilities

A fictional two-stage token tree illustrates next-token prediction.

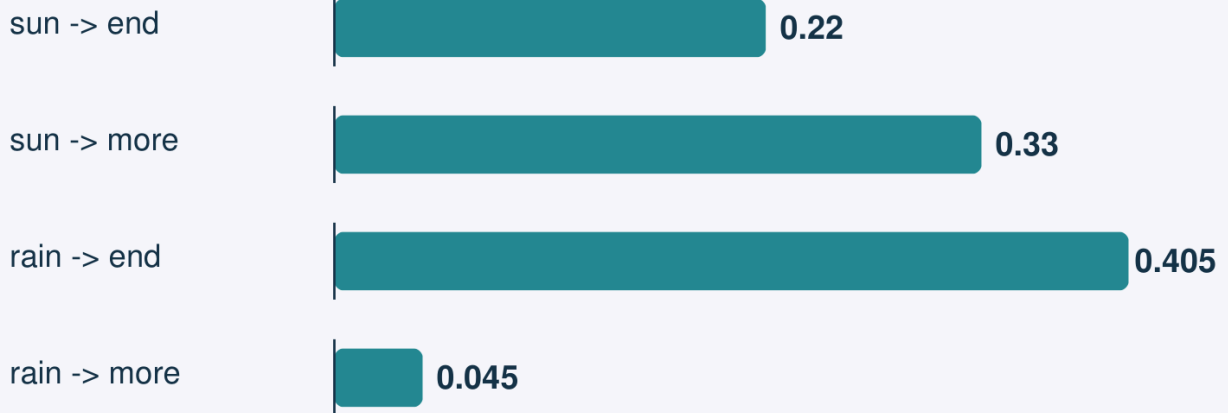
The facts we will use

First token probabilities: sun 0.55 , rain 0.45 . Given sun: end 0.4 , more 0.6 . Given rain: end 0.9 , more 0.1 .

These are complete toy branches; end paths stop, more paths do not.

Joint probabilities of the four toy branches

probability



Bar length shows the supplied value; the marked baseline is zero.

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end: $0.55 \times 0.4 = 0.22$; sun -> more: $0.55 \times 0.6 = 0.33$.



3. Rain branches

rain -> end: $0.45 \times 0.9 = 0.405$; rain -> more: $0.45 \times 0.1 = 0.045$.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. First distribution

What happens and why: sun 0.55, rain 0.45: greedy first choice is sun.

Step: 2. Sun branches

What happens and why: sun → end: $0.55 \times 0.4 = 0.22$; sun → more: $0.55 \times 0.6 = 0.33$.

Step: 3. Rain branches

What happens and why: rain→end: $0.45 \times 0.9 = 0.405$; rain→more: $0.45 \times 0.1 = 0.045$.

Step: 4. Compare end paths

What happens and why: rain→end is the more probable of the two complete end paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

What this example does not establish

These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: The model always chooses the single most likely continuation. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Always report the prefix, alternatives and stage. The complement of ready in the first table is busy or unknown, probability 0.5; it is not automatically busy. Values rounded for display may not sum exactly to 1, so retain full values for calculations and label rounding. A missing or invalid table should lead to an explicit error, not a confident probability claim.

Exit: explain two errors in “ready has probability 0.5, so busy also has probability 0.5 and the room is ready half the time”.

Apply the connected idea: Embeddings and position

State whether a vector is a token embedding, a position vector or their combination before calculating distances. Position changes can affect numerical similarity even for the same token. An unknown token or a sequence beyond the toy position table needs explicit handling. This two-position contract rejects a third position; it does not silently wrap back to position 0.

Exit: a program compares vocabulary IDs instead of vectors and silently repeats its two-position table for a longer sequence. Identify both contract errors.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported table: first red=0.4, blue=0.6. End given red=0.9 and end given blue=0.5. The red,end path has probability 0.36. The blue,end path has probability 0.3 despite blue being the larger first-step probability.

Guided: compute both products and distinguish the preferred first token from the more probable specified end path.

Apply the connected idea: Embeddings and position

Supported case: embeddings $A=[2,1]$, $B=[1,3]$, positions $[0,0]$ and $[1,-1]$. Sequence A B becomes $[2,1]$ and $[2,2]$. Sequence B A becomes $[1,3]$ and $[3,0]$. Both sums are $[4,3]$.

Guided: compute both ordered representations and compare their sums. Explain which operation discarded the difference.

Practise with less help: Calculate the paths

Use workbook W2 for the connected example “Next token distributions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Next token distributions. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Embeddings and position

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A position vector guarantees every summary preserves order

Explain the picture

Why is 0.625 not a broken probability sum?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Build a weighted token spinner, sample twenty times, tally outcomes, and compare observed counts with expected relative weights.

Build a weighted token spinner, sample twenty times, tally outcomes, and compare observed counts with expected relative weights.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Explain that Study Buddy responses come from possibilities, so important claims need checks.

THIS WEEK'S CHECKPOINT Explain that Study Buddy responses come from possibilities, so important claims need checks.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Run a weighted-word simulation and compare repeated outputs.

CHALLENGE Run a weighted-word simulation and compare repeated outputs.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A toy next-token distribution is rain .5, sun .3, snow .2.

The question

Check total and interpret the largest value.

Worked explanation

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A toy next-token probability distribution

Evidence or stage · Value or result

Evidence or stage: rain

Value or result: 50

Evidence or stage: sun

Value or result: 30

Evidence or stage: snow

Value or result: 20

A toy next-token distribution is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

14. Try a new case independently

Independent case: Analyse a new table independently

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum. Which first token is greedy, which end path is largest, and does either verify a real status?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A toy next-token distribution assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Explain that Study Buddy responses come from possibilities, so important claims need checks.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A toy next-token distribution is $\text{sun}=.5$, $\text{rain}=.3$, $\text{snow}=.2$. Over ten samples, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.